

Secondary city regionalism: China's cross boundary development projects

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ABSTRACT

This study examines the modalities of recent city regionalism, initiated by secondary cities, in post-reform China. Academic and policy discourse has centred on the emergence of, and state rescaling for, global city regions. However, the role and strategizing of secondary cities within their respective city regions have often been overlooked. This study fills a gap by examining city regionalism from the 'outside in', focusing on secondary city authorities' strategies to reap the benefits of agglomeration and buffer the effects of marginalization by mobilizing cross boundary development opportunities. Based on fieldwork, we identified three types of cross boundary developments: new towns, industrial parks, and innovation enclaves. The regionalization strategies and partnership programs launched by secondary cities highlight how they identify and exploit niches within city regions. This article notes how future research can pay more attention to the agency of secondary cities and the emergence and characteristics of inter-urban places within city regions.

1. Introduction

The expanding scale of urbanization, processes of urban economic agglomeration, and purposeful state strategies centred on poly or multi-city regions has attracted much recent attention (Brenner, 2014; Harrison & Hoyler, 2015; Laquian, 2005; Scott, 2001). Much of this literature treats city regions as undifferentiated units of analysis: scaling up an appreciation of economic and policy dynamics from the city to the city region to the polycentric city region (Yeh & Chen, 2020); rarely reading, let alone theorizing, the urban system, economic or policy landscape from the 'outside in' (Keil, 2014; Phelps, 2021a); or failing to position secondary cities within intra-city region divisions of labour and attendant policy imaginaries suggested by the literature on urban polycentricity (Hall & Pain, 2012; Phelps & Ozawa, 2003; Rodríguez-Pose, 2008; Yang et al., 2024).

The literature on city region formation has several strands. A first strand of literature focuses on the economic forces that generate the largest of global urban giants, with particular reference to the logic of agglomeration, growth, and spatial interactions (e.g., Derudder et al., 2022; Scott, 2001) and is especially embedded in Western success stories (Kantor & Nelles, 2015). A second strand of literature has focused on mapping the polycentricity of global city regions based on an advanced producer-services economy and information connectivity (Reades &

Smith, 2014) contributing to debate on the division of labour between the constituent cities (Meijers & Burger, 2017; Volkmann & Rusche, 2020). A third strand of literature seeks to unpack the political and governance dynamics of city regions (e.g., Herrschel & Newman, 2002; Jonas & Ward, 2007; Lang & Knox, 2009; Li & Wu, 2012; Raiser & Volkmann, 2007; Tewdwr-Jones & McNeill, 2000).

Yet much remains to be understood of the discursive and policy construction of city regions (Harrison & Gu, 2021; Hincks et al., 2017). It is important to distinguish between ex-post urban and economic planning after the formation of city regions and ex-ante envisioning contributing to the formation of city regions (Harrison & Gu, 2021). The existing literature on urban and economic planning of city regions has tended to focus on the strategies of nation-states and primary city authorities with little attention paid to secondary cities, where secondary cities are defined in a simple way as all non-primary cities within individual city regions (Cardoso, 2016, 2018). In this paper, to suit the Chinese context of urban administrative hierarchies (Ma, 2005), we refer to secondary cities as: (1) lower-tier prefectural-level cities excluded from centrally administered municipalities (Beijing, Shanghai, Tianjin, Chongqing) and provincial capitals (such as Nanjing of Jiangsu province); (2) economically significant county-level cities that have economic scale and specialized roles in regional (e.g. the Yangtze River Delta Region) value chains (e.g. Kunshan), and; (3) cities of the same

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administrative status but of much lesser scale (e.g. Chuzhou, Anhui) than their prefectural-level city counterparts (e.g. Suzhou, Jiangsu). As we seek to show, these cities appear to be excluded from, or are on the margins of, city-regionalism orchestrated by administratively or economically-primary cities. Yet their governments have sought actively to reshape city-region agglomerative forcefields to their advantage.

This article seeks to fill this conceptual and empirical gap. It poses the following question: how have secondary cities responded to agglomeration-led processes and what urban and economic strategic planning imaginaries and practices have been apparent with respect to city regionalism? In answering this question, we make two contributions to the literature. First, we focus on the strategic struggles of secondary city authorities to find niches in the economic organization and governance arrangements of city regions. Second, and emerging from the China setting for this study, the paper draws attention to the mobilization of cross-boundary development opportunities as part of secondary city authority strategizing within city regionalism.

We begin in the next section by charting a path through the extant literature towards an understanding of the key elements of secondary city regionalism as it has emerged in Chinese city regions. We then outline the methods used to collect and analyze the documentary and interview-based data on which this paper is based. We go on to present findings on the way in which secondary cities have developed cross-boundary new towns, industry parks and innovation enclaves as a means of leveraging economic spillovers present across global city regions by virtue of borrowed size and function effects. We conclude with suggestions for future research to focus on: the strategic urban and economic planning agency apparent among city authorities across city regions, including secondary cities particularly; the role of cross-boundary development projects in the shaping of inter-urban places and economies within city regions.

2. Secondary city strategizing in global city regions

Viewed from afar, the promotion of city regions follows a logic of urban economic agglomeration, trickle-down spillovers and associated urban planning focused on primary cities (Jonas, 2013) within national territories and even the historic cores within those primary cities (Newman & Thornley, 2005). This logic and rhetoric of primary city agglomeration to global city regionalism not only privileges but also expands primary city spheres of economic influence (Beel et al., 2016). In China, the growing intra-regional disparities between Beijing and its surrounding areas embody the consequences of such a spatially selective, primary city-first-agglomeration approach to city regionalism (Li & Jonas, 2023).

The dominant critique against agglomeration-led city regionalism is based on the absence of the trickle-down effects from and within city regions often implied in global city region imaginaries in particular (MacKinnon, 2021) and indeed the negative impacts on a range of provincial towns and rural areas beyond global city-region hinterlands (Pemberton & Shaw, 2012). However, the impact on the periphery *within* city regions, and the resulting responses of secondary cities remain underexplored. In this study, we go further to explore the agency and strategizing of secondary cities to provide an outside-in rather than an inside-out perspective on city region development. We argue that alongside the nation-state and private sectors, secondary city governments are promoters of city-regional imaginaries.

The material processes of city regional formation are subject to strategic spatial planning in which greater or lesser imagination is exerted not just by stakeholders aligned with primary cities (Healey, 2006) but by a multitude of stakeholders including the governments of constituent primary and secondary cities. Thus, it is suggested that city regions are not purely economic outcomes, but rather constantly evolving political-economic projects (Harrison & Gu, 2021). Indeed, city region formation is full of politics (Schafran, 2014; Wachsmuth, 2017), nowhere more so than in China. In China, the imaginaries of secondary

cities have sought to leverage and divert agglomeration effects by borrowing size and function from, and averting the shadow effects of, primary cities. Our study highlights the urban and economic strategic planning imagination that secondary cities have adopted in the tactical construction of cross-boundary development projects.

2.1. Secondary cities and global city region divisions of labour

The literature on urban polycentricity alerts us to several facets of city region economic and policy dynamics that are germane. First, the term itself alerts us to the multiple urban centres and economies that comprise city regions. Second, while three different modes of city region formation could be discerned (Champion, 2001), none of the ‘centrifugal’, ‘incorporation’ or ‘fusion’ modes may quite do justice to the strategizing implicated in, and effects of, secondary city urbanization and economic development. Specifically, these modes of incorporation are suggestive of the urban morphological dimensions of polycentricity and the imperfectly associated constraints, on and opportunities open to, secondary cities at their jurisdictional boundaries within city regions. Third, however, polycentricity can be functional and accompanied by a division of labour among the different urban economic agglomerations that make up city regions, providing further specific foci for secondary city authority strategizing including with respect to leverage economic opportunity across jurisdictional boundaries specifically.

We can elaborate on the division of labour within city regions with reference to a more specialized but partially overlapping literature on borrowed size (Meijers & Burger, 2017; Phelps et al., 2001; Phelps & Ozawa, 2003). Alonso (1973: 200) first identified borrowed size as a feature ‘whereby a small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population centres’. Borrowed size can be summarized in terms of the opportunities for businesses to benefit simultaneously from location in smaller settlements, with less congestion and lower rents, and access to larger settlements, with sizeable markets, business services, large and diversified labour markets (Phelps et al., 2001) as a feature of the polycentric city regions (Phelps & Ozawa, 2003). The advantages that secondary cities might possess in terms of borrowed size have since been distinguished from the advantages of borrowed function. These advantages can, in turn, be distinguished from the shadow effects whereby settlements in the orbit of larger settlements may be subject to negative externalities (Meijers & Burger, 2017). And indeed, borrowed size effects have been identified by one recent study as contributing to the performance of small and medium sized city economies in the Yangtze River Delta Region (Zhao et al., 2025).

2.2. Secondary city strategizing of cross-border projects: a framework

Despite increased recognition of the significance of urban peripheries – suburban communities and secondary cities – within city and megalopolitan-regions (Keil, 2017; Kruse & Sugrue, 2006; Phelps, 2010; Phelps & Wu, 2011) including those in China (Cheng & Shaw, 2018; Li et al., 2025; Wu & Phelps, 2011) and the need to move beyond ‘methodological cityism’ (Brenner, 2014; Keil, 2014; Phelps, 2021a), much regarding the ramifications of the scale and complexity of diverse urban peripheries remains to be elaborated. Indeed, the difficulties of adequately characterizing and bounding urban peripheries outside of China have been noted (Phelps et al., 2010) with new urbanization in the United States being considered edgeless or unbounded (Lang, 2003). One of the least researched issues in this respect is the extent, nature and effects of political-bureaucratic and economic strategizing focused on jurisdictional boundaries – and, as we seek to elaborate, China is *the* case in point.

Boundary objects are commonly considered residual artifacts of classification schemes (Bowker & Star, 2000) and, by extension, the value of boundaries and boundary objects is neglected in the urban planning literature (Phelps, 2025). Yet one concrete means by which

secondary city authorities' attempt to construct borrowed size and function advantages within city and megalopolitan-region economic and political divisions of labour is the planned development of new towns, industry parks and special economic zones at the boundary of their jurisdictions. These cross-boundary development projects are notable for their explicit fueling of an inter-urban or 'interplace' economy within national city systems and city regions (Phelps, 2004, 2017, 2021b). In terms of urban and economic planning, boundary objects manifest negatively in the form of locally unwanted land uses as a result of the dumping of externalities upon neighboring jurisdictions. However, historically and presently urban peripheries either side of jurisdictional boundaries commonly have also been spaces in which the imagination of citizens and private developers has been focused, contributing to extensive patterns of urbanization (Phelps et al., 2025).

The notable feature in China is how much of this imaginative attention on urban peripheries has also been exercised by municipal governments and secondary city governments in particular. Moreover, this strategizing by local governments connects to the interests of private developers as part of China's state entrepreneurialism (Wu, 2020), and to citizens for whom suburbanism as a way of life is understood less in terms of a bourgeois retreat into distinct suburban communities with autonomous governments than as an approach to the city based on location choices regarding access to housing, services and transportation (Ren, 2021). These shared interests of the state, private sector and citizens manifest in the strategizing, development and consumption of boundary objects in the form of cross-border new towns, industry parks and innovation zones.

The strategizing of cross-boundary new towns sits at the state end of a spectrum of state entrepreneurship arrangements since it typically requires state development corporation-led annexation and conversion of rural land (by definition at the periphery of city boundaries), and the planning and servicing of land for development. The secondary city strategizing logic is one of attracting economic and population spillovers (Li & Yuan, 2021) - in some instances seeking to level-up to primary city service standards and in other instances seeking to 'borrow' urban functions. As diverse and self-contained new settlements, new districts are not economic enclaves but contribute significantly to urban coalescence within city regions.

Other cross boundary developments in the form of industry and innovation park special economic zones (SEZs) can resemble economic enclaves as a result of their logic of attracting businesses and talent (Li et al., 2023; Phelps et al., 2023). However, they also are subject to extensive incentivizing and strategizing by all tiers of government as part of desires for balanced urban and regional development and can leverage emergent industry divisions of labour and spillovers evident at urban and regional scales.

Cross-boundary industrial parks refer to new industrial spaces built in secondary cities in cooperation with the primary city authorities to borrow economic size, triggering a trickle-down of industry and manufacturing from the cores to the peripheries of city regions (Zhang et al., 2020). They are numerous standardized developments catering to the continued industrialization of secondary city economies, land price differentials and the strategized redistribution of industry from primary cities (Boyang et al., 2014).

Cross-boundary innovation enclaves – innovation parks – refer to the spaces deliberately created by secondary cities positioned within global city regions to borrow from large and diverse labour markets, leveraging on the agglomeration of talent and technology. They represent a strategic move by city governments – including secondary city governments – to leverage emergent divisions of high-tech labour and technological spillovers. Cross boundary innovation zones often have been superimposed upon or carved from cross-boundary industry parks and new towns as part of the evolution of SEZ policy (Miao & Phelps, 2022).

This categorization of boundary objects and associated cross-boundary imaginaries aims to articulate the underlying actors and their different contexts and motives. Emphasizing agency and incentives

helps unpack how primary city agglomerative forcefields are shaped and leveraged in practice. This paper argues that the agglomeration and trickle-down of city regions do not operate like a natural law but are shaped by the shared interests among state, development and other private sector actors and citizens. The emphasis on the government and private sector actors of the periphery of city regions helps to extend existing knowledge, which has mainly been acquired through the perspective of primary cities and the actors therein.

3. Definition, data and methods of analysis

The term "city region" is a fuzzy concept (MacLeod & Jones, 2007). In this study, a city region refers to a cluster of interconnected urban settlements, anchored by one or more large cities. As a result of extensive annexation, most city regions including those surrounding the global cities of Beijing and Shanghai but also the likes of Hangzhou and Suzhou in China are themselves large but also fit within megalopolitan regions, such as the Beijing-Tianjin-Hebei (BTH) and the Yangtze River Delta (YRD). The scope for economic and policy differentiation within such city or megalopolitan regions is considerable given their incorporation of numerous second and lower tier cities (Zhang & Wu, 2006).

After years of relative neglect, increasing attention has been paid to the economic and policy dynamics of secondary cities (Cardoso, 2016, 2018). Even so, many of the specifics of how secondary cities strategize remain poorly understood. This study uses the term secondary cities, in a basic way to refer to non-primary cities within individual global city regions. The classification of these cities is loosely defined to represent the development dilemmas faced by these cities, whether they attempt to subvert their second-tier status or remain in the shadow of their larger counterparts. The aim of this study was not to provide a meticulous framework for categorizing small, medium, and large cities within a regional, national, or international urban system. Instead, examined through a relational lens, we seek to demonstrate the concrete manifestation of strategizing of secondary city authorities in a context of uneven development in global city regions.

These definitional and methodological challenges are amplified given the difficulties of defining urban peripherality, and hence urban peripheral boundary objects. In the Chinese urban system - which distinguishes only between urban and rural – academics and practitioners are left to engage in 'boundary work' when somewhat arbitrarily identifying settlement space that is the functional equivalent of western suburbs. Of relevance to our study is the fact that new towns are considered the nearest equivalent to suburbs or urban peripheries in Chinese urban planning practice (Ren, 2021).

Our study is based on more than 20 field visits to China's city regions, mainly the BTH and YRD, between 2008 and 2023. Repeated field visits allowed us to develop sensitivity to the diffuse network of actors underlying city-regional development and also provided opportunities to build trust for conducting interviews and collecting internal documents and other secondary materials. Over ten sites were visited with some more than once, then cases were selected among them to represent the three different boundary object planning instances emphasized in this paper (Table 1). Interviews were conducted with various key figures involved in the policymaking process, and interviewees were carefully selected using a snowball sampling strategy. To triangulate the interview data, efforts were made to broaden the range of interviewees to include local officials at the provincial, municipal, and county/district levels; planners; university academics; entrepreneurs; and residents. In total, fifteen interviews with government, private enterprises, planners and other informants were conducted. Six government symposia regarding the small cities and towns development were also attended. The interviews varied from 30 min to over 1 h and together with the government symposia totaled approximately 50 h. Transcripts were made immediately to ensure the accuracy of recollection. To ensure confidentiality, all names of the interviewees and individuals mentioned by the interviewees were anonymized. A wide range of secondary data

Table 1
selection of case and interviewees.

Interview	Date	Examples	Relevance
1	15 June 2020	- Gu'an New Town	Real estate professional
2	20 April 2021		Enterprise
3	9 March 2010	- Huaqiao New Town	Chief planner
4	11 April 2013		Government official
5	31 March 2023		Government official
6	21 October 2021	- Jiangyin-Jinjiang industrial park	Enterprise
7	16 August 2022		Government official
8	18 November, 2021	- Suzhou-Chuzhou industrial park	Enterprise a
9	18 November 2021		Enterprise b
10	19 November 2021		Government official
11	10 November 2022	- Changxing innovation enclave (Shanghai)	Government official
12	22 July, 2024		Enterprise
13	23 July 2024		Enclave official
14	13 December 2021	- Quzhou innovation enclave (Hangzhou)	Enclave official
15	14 December 2021		Enterprise

was also consulted, including local government websites, local newspapers, and published works.

Alongside the review of documents, we believe the breadth and depth of these interviews allows us to comment on the basic characteristics of cross-boundary development types and their motivation.

4. The mobilization of cross-boundary developments by secondary city governments

In this section we provide empirical evidence of secondary cities' attempts to better position their economic fortunes within city regions. These attempts focus on jurisdictional boundaries and development projects that are important constituents of the specifically inter-urban economy of national city systems and city regions (Phelps, 2004, 2017). Each type of development project is illustrated with two or three examples from either BTH or YRD to show the logic and concrete measures (Table 2). The locations of these examples are marked in Fig. 1 to demonstrate the geography and spatial relationships of the cases.

4.1. Cross-boundary new towns: borrowing function

The cross-boundary project of new towns is derived from the dominant suburban spaces emerging in China, which are characterized by formal, master-planned, and state-led spatial production (Shen & Wu, 2012). The development of new towns has been led by China's well-known municipalities, such as Guangzhou, Shanghai, and, more recently, Beijing. These new towns take a variety of forms, ranging from development zones to university towns and residential suburbs, planned or built by different levels of government, including central, provincial, and county governments and their development corporations (Li et al., 2020). According to Guangzhou's latest plans, 29 new towns have recently been built or will soon be built (Li et al., 2020). Shanghai has

recently updated its new town plan based on the earlier “one city, nine towns” (China Daily, 2021). Beijing, China's national capital, has developed Shunyi New Town, which is becoming an employment hub, and has recently announced that it will build Tongzhou as a new sub-center to relocate some economic functions and relieve congestion in the central city (Wu & Phelps, 2011; Zou & Zhao, 2018). New town development is pervasive as a means of capital accumulation for the local state and facilitated by booming property markets (Shen & Wu, 2017).

Witnessing the vibrant economic growth of its primary city counterpart, cross-boundary new towns are imagined by these secondary cities as outer suburban spaces of primary cities. In other words, the project of cross-boundary new towns is articulated by secondary cities to physically and functionally integrate into neighboring large cities. As a result, measures have been taken to reduce border barriers and facilitate the spatial extension of trickle-down effect of metropolitan agglomeration or reduce agglomeration shadow effects. There are two aspects of these measures. The first is branding proximity to the metropolis, facilitated by convenient regional transport, such as in the case of Gu'an New Town (Li & Yuan, 2021). Gu'an is a county-level city in the Langfang Prefecture of Hebei Province, on the Beijing-Kaifeng Expressway. Gu'an's proximity to Beijing's Tian'anmen Square, is used to promote its unique location, as is clear from a simple internet search or an examination of on-site billboards (Fig. 2). Thus, Gu'an is positioned under a clear city-region vision, and not promoted as a jurisdiction under Langfang—a mediocre city in the Hebei Province. This positioning of Gu'an under the great metropolis of Beijing draws attention to its locational advantages, and helps attract inward investment and spillover industries from Beijing. Similar cases can be seen in Huaqiao, Kunshan in Jiangsu provinces, which are close to Shanghai. The first trans-provincial metro line in China was built to integrate the new town of Huaqiao into Shanghai (Li & Wu, 2018).

The second aspect is to focus on new towns as a means to borrow or

Table 2
Type, strategizing and exemplars of cross-boundary developments.

Development projects	Strategizing in the city region	Examples	Background information
Cross-boundary new towns	- Borrowing function - primary city brand and service standards	- Gu'an New Town - Huaqiao New Town	- A new town developed at the edge of Langfang, with close distance to the metropolis Beijing - A new town developed at the edge of Kunshan with close distance to the metropolis Shanghai
Cross-boundary industrial park	- Borrowing size (mass) of industries	- Jiangyin-Jinjiang industrial park - Suzhou-Chuzhou industrial park	- A joint industrial park developed by Jiangyin and Jingjiang, Jiangsu province, here Jinjiang as the secondary city - A joint industrial park developed by Suzhou (Jiangsu) and Chuzhou (Anhui), here Chuzhou as the secondary city
Cross-boundary innovation enclaves	- Borrowing size of high-tech talents, borrowing function – industry divisions of labour	- Changxing innovation enclave (Shanghai) - Quzhou innovation enclave (Hangzhou)	- Innovation enclave developed by Changxing (Zhejiang province), the secondary city, in the metropolis of Shanghai - Innovation enclave developed by Quzhou (Zhejiang province), the secondary city, in the metropolis of Hangzhou

Source: the authors.

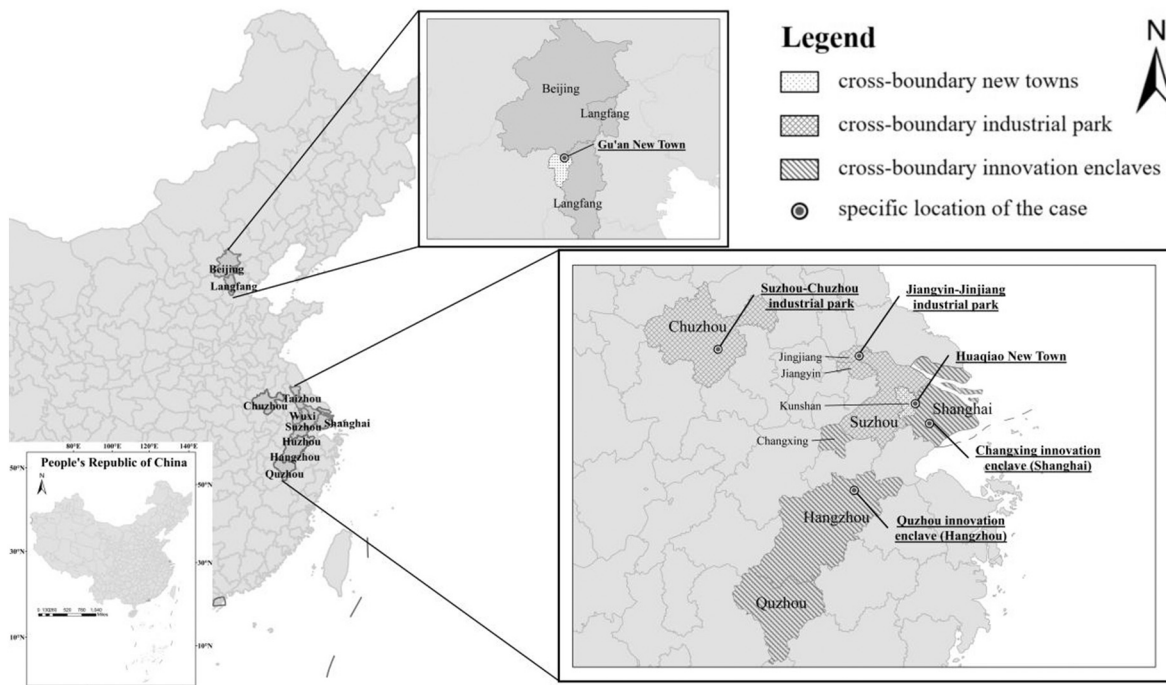


Fig. 1. Locations of cross-Boundary development exemplars.



Fig. 2. A billboard advertisement of Gu'an.

Source: Reproduced from H&H, <http://www.huayuhua.com/index/Anli/show/catid/7/id/56.html#>, accessed 1 September 2020.

equalize the urban functions or service standards of the new town and nearby primary city. In the case of Gu'an, the idea was to improve collective services and standards inside and outside the new town to match those of Beijing or its suburbs (interview, 20 April 2021). These measures are not limited to urban planning and beautification but also include healthcare and education. Together, they aim to build four dimensions – charisma, attractiveness, capacity, and competitiveness – to attract high-end industries (Li & Yuan, 2021). This signals to investors that they can enjoy Beijing's infrastructure, including airports, amenities, education, healthcare, scientific research, and talent while benefiting from Gu'an's low production and reproduction costs. The

concentration of resources in a new town often creates a stark contrast with the suburbs of the nearest metropolis. One local planner, based in Shanghai, commented that the high-rise buildings in the neighboring new town (Huaqiao, Kunshan) confuse people about Shanghai and Kunshan's locations. Common belief might be that Shanghai is more developed, and that its suburbs would be better in most respects. However, the reality is very different, that is, the new towns of these neighboring secondary cities are much more modern and better equipped, with more public facilities than those of Shanghai (Interview, 9 March 2010).

It is unclear how far the purposeful strategizing by secondary cities to 'pull' population and industry has shaped the formation of city regions as a whole when compared to the 'push' of negative externalities of high land and property prices and salaries associated with primary cities. Population spillovers are particularly pronounced in the cross-boundary new towns adjacent to Beijing. As the 2022 Annual Report on Beijing Commuting shows:

"About 330,000 people commute to Beijing from the surrounding areas every day, accounting for 3% of the total number of commuters in Beijing, mainly distributed in Langfang, Gu'an, Zhuozhou and other places... The destinations of commuters from Langfang are mainly the CBD and Wangjing, the downtown of Beijing, and Yizhuang, a suburb of Beijing, and the destinations of commuters from Gu'an are mainly Fengtai Science and Technology Park."

(Liu, 2023)

A similar situation can be observed between Shanghai and the nearby cross-boundary new town in Kunshan and Taicang. According to the 2020 Annual Report on Cross-Boundary Commuting in the Yangtze River Delta, Kunshan and Taicang are the main sources of commuters flowing into Shanghai, accounting for 74.8% and 12.7%, respectively. In addition to the nominal increase in population spillover, industrial links were also fostered in these areas. The authors were able to gain access to results from a survey that is not publicly available during fieldwork on location. According to a survey conducted by Taicang in 2022, a county-level city in Suzhou, Jiangsu Province, near Shanghai, of the 953 enterprises that participated, 671 were linked to Shanghai upstream or downstream of the industrial chain, accounting for 70.4%; this

percentage was even higher in manufacturing, where 76.9% was related to Shanghai within the industrial chain; in comparison, only 50% of the service sector enterprises are linked to Shanghai, 60% of which are in trade and logistics (interview, 31 March 2023).

In summary, one particular strategy of secondary cities has been to develop new towns at a short distance to the center of the metropolis, divided by an administrative boundary that separates the two neighboring cities. Transport infrastructure investment and leveling tactics were used to reduce the border-related barriers that otherwise would impede the space of economic flows. These measures have significantly improved the mobility of population and industrial links. In particular, the incentive for a neighboring new town to attract the population is a means rather than an end. This is because China does not have a property (housing) tax but is based on production (manufacturing) tax. As a result, the property boom benefits the government only in the form of a one-time sale of residential land. Therefore, local governments use revenue from land sales to fund industrial attractions, creating long-term revenue streams. It is hoped that the population spillover will ideally match the industrial spillover; that is, the people attracted to the new town will also work in the industries they attract. However, in reality, they often do not match. According to the Taicang survey, 623 of the 953 companies surveyed had employees who commuted, accounting for 65%, of whom 54% commuted daily from Shanghai to Taicang; further, 29.6% of the commuters were managers, and 37.3% were technicians. Overall, the cross-boundary new town strategy increases the economic flows between secondary cities and the metropolis, though the familiar challenges of new towns achieving a high measure of self-containment remain.

4.2. Cross-boundary industrial parks: borrowing industry mass

Cross-boundary industrial parks denote the motivation of secondary city jurisdictions to generate local economic growth, by producing spillover attractions from the metropolis. These secondary cities are generally located further from the metropolis than is the case of those developing cross-boundary new towns, and therefore, less prone to attract investment from foreign capital. In addition, without the initial industrialization, such as in town and village enterprises (TVEs), these secondary cities missed the first opportunities for development, at the beginning of economic reform. Against this background, cross-boundary industrial park proposals are revealing of a secondary city-regional imaginary. These imaginaries are greatly facilitated by the pioneering development of leading metropolises, and the successful progress in regional transport in China.

The first initiative can be traced back to the Jiangyin-Jingjiang Industrial Park, established in 2003 in Jingjiang, a neighboring city of Jiangyin, and divided by the Yangtze River (Luo & Shen, 2009). This imagination was made possible only by the completion of the national highway and bridge connecting Jiangyin, and Jingjiang across the Yangtze River. Prior to this, Jiangyin was highly developed, with economic reform through TVEs and their pioneering shareholding reforms. Jingjiang, on the other hand, was underdeveloped because of its disadvantageous transport location, although it has natural port conditions. The city-regional imaginary was shaped by joint industrial parks in which both jurisdictions were deeply involved. Initially, its materialization heavily relied on the intervention of a higher level of government, to facilitate communication between these two jurisdictions given the unequal power dynamics (Interview, 16 August 2022). Authorities of the more economically developed jurisdiction have less incentive to cooperate compared to those of a lagging jurisdiction. With the guidance of the provincial government of Jiangsu, the first phase of the registered capital of the joint industrial park was set at 100 million yuan—Jiangyin 90%, Jingjiang 10%—and 10 years of investment income remained in the park's rolling development. In other words, under the intervention of the provincial government, the city-regional imaginary was built not only around economic complementarities, but also using the active

redistribution of industry. It is hoped that the use of land and port resources in Jingjiang, and the relocation of industries from Jiangyin to Jingjiang, will benefit regional development. Unexpectedly, however, the economic situations of these two cities underwent dramatic changes that had a negative impact on cooperation and city-regional development (Interview, 16 August 2022). On the one hand, Jiangyin's economy was gradually badly affected by the influx of FDI. On the other hand, Jingjiang began its period of development independently, accusing Jiangyin of merely relocating unwanted industries and using too much of its land resources. Consequently, the city-regional initiatives stalled after the first decade, and made little progress. In summary, the local government's attempt to borrow size of industries and attract manufacturing spillovers, by relocating companies from one city to another, proved to be fraught with conflict. First, the cost of relocation for firms was very high, discouraging them and increasing administrative costs for local governments. Second, the lack of progress in attracting businesses and the consequent slow growth in revenue put more pressure on the management of the joint park after a few years when it was time to share the profits.

Because of these failures, secondary cities began to focus on borrowing size by attracting production expansions of existing industries, rather than enterprise relocation. The case of the Suzhou-Chuzhou Industrial Park vividly illustrates this phenomenon. The Suzhou-Chuzhou Industrial Park is a project where the Sino-Singapore Suzhou Industrial Park Development Group Co., Ltd. (hereafter, Sino-Singapore Park) was invited to work with the Chuzhou government, to develop a joint industrial park. Unlike previous initiatives, the city-regional imaginary did not directly involve the local governments of the more developed municipalities. Instead, due to the well-documented unique international reach and administrative status of the Sino-Singapore Park authority in particular (Miao, 2018; Pereira, 2003), the strategizing has been implemented indirectly via the respective development zone authorities. As an interviewee explained:

"70–80 percent of the companies attracted is from the Yangtze River Delta, many of them are small private and foreign companies, not state-owned enterprises, they came to our park because of the Sino-Singapore Park brand. Initially, the Sino-Singapore Park took full responsibility for attracting investment. But it was very slow. Since 2015, the Chuzhou Management Committee began to attract companies. Two-thirds of these companies attracted to Chuzhou was done by the the Chuzhou Management Committee. Nevertheless, we shared business attraction agency fees with Sino-Singapore Park. [...] This is because we used their brand. This resulted in a significant difference. [...] Many of the companies come here because of Sino-Singapore Park management, not Chuzhou itself. Especially for those with some sort of high technology, four-fifths of the investment has been attracted by Sino-Singapore Park. The relative attractiveness of Sino-Singapore Park and Chuzhou is very different."

(Interview, 19 November, 2021)

The later city-regional imaginary was, thus clearly not built on the relocation of production but on the spillover of brands and services from the metropolis, Suzhou, to the secondary city, Chuzhou. To achieve the off-site replication of the park model, it is necessary to implement standardized chain management, and develop headquarters and sub-parks as a whole, so that companies can see the same quality and enjoy the same services in each park. Here, it is perhaps even harder to disentangle the effects of secondary city strategizing given its indirect exertion via zone authorities when set against the importance of corporate drivers of innovation linkages across the entire Yangtze River Delta Region (Li & Phelps, 2019) and investment and ownership linkages between zones in the Suzhou-Changzhou-Wuxi metropolitan area (Zhang, 2022).

In conclusion, the imaginary of a cross-boundary industrial park is built on industry expansions and supply chain flows or transactions to borrow size but is also reliant on the borrowing of function. On the one

hand, the flows include communication between (quasi-)public sectors. The head office of a successful development zone exports its brand name and receives a certain amount of management fee for using the brand. In return, management services are exported to the joint park in the process of cross-boundary cooperation, to improve the business environment of the secondary cities in lagging areas. For example, Suzhou provides incubator training and other innovative services to the Suzhou-Chuzhou joint park. Thus, the two sides of government have realized interactions in light of exchanging information, cadres, business resources and so on. On the other hand, the flows also include supply chains between the companies located in the joint park and in the original host cities. This is because many companies in the original park are subsidized by the joint park to set up branch factories, when they need to expand their production capacities. Overall, the imaginary of cross-boundary industrial parks facilitates spillovers by generating invisible flows, rather than physical relocations.

4.3. Cross-boundary innovation enclaves: harnessing large talent pools

Secondary cities want to attract not only production and population spillovers but also produce services and talent, which arguably are associated with greater local added value and income. In fact, for both the cross-boundary new towns documented in the previous section in Gu'an, near Beijing, and Huaqiao, near Shanghai, residential real estate was built not only as typical residential suburbs, but also imagined and developed with the view to attract the talent to develop local industries. Unfortunately, attracting talent to settle in peripheral cities is challenging in China (Miao, 2017). The context is very different from edge cities in the US, where major concentrations of white-collar and research and development employment specifically is common in outer suburbs (Garreau, 1991).

Cross-boundary innovation enclaves emerged across the YRD region, within this context. From 2015 to the end of 2023, 22 innovation enclaves were established in Shanghai and other cities in the YRD region (interview, 10 November 2022). This number is higher in Hangzhou, where the Zhejiang Provincial Government took a more active role in encouraging secondary cities to establish enclaves in the provincial capital, as a strategy for high-quality development. Compared with cross-boundary industrial parks, where secondary cities tend to offer a large area to attract manufacturing, innovation enclaves set up by secondary cities in the metropolis often cover a small area, such as the rental a few floors of office space. These enclaves operate as incubators and innovation platforms that focus only on a few high-tech industries. Targeted industries are often strategic and emerging ones, such as artificial intelligence, machinery and equipment, new energy and high-end manufacturing, and medical and pharmaceutical industries. The hope is that, at a later stage, when technologies and businesses are mature, they will locate their factories in, and hire workers from, secondary cities. In other words, the motivation for secondary cities to establish an innovation enclave is to harness the size of talent and associated industry secrets or technological knowledge in the metropolis. An interview with a Changxing government official confirmed how the establishment of the first cross-boundary innovation enclave echoed this view:

“The context of the imaginary was that Changxing faced a bottleneck in attracting businesses. Every year, the county faced great challenges and made bigger efforts to attract enterprises but with poor results. It is difficult to attract more advanced industries, whereas traditional local industries find it difficult to upgrade themselves. Under the huge pressure of development evaluation, local officials have begun to look for a way out. They know that they cannot compete with leading cities that have more political and financial resources. This means that they are very disadvantaged in attracting giant enterprises that are very competitive, with the preferential policies that county-level governments do not have fiscal and regulatory power to offer. Therefore, they began to consider small, high-

tech companies and start-ups. As a result, officials began to survey local high-tech and small businesses, and the opinions gathered confirmed that future-oriented technology should be the target industry. Officials began to consider how to provide space to attract technology. They realized that the space had to be close to Zhejiang University in Hangzhou, the top university in Zhejiang. Coincidentally, Changxing's county chief and party secretary used to be Hangzhou's district chief [i.e., all with links to Hangzhou], which made it easier for Changxing to rent the wasteland and start incubating the enclave.”

(Interview, 23 July, 2024)

Rather than creating a local economic space and facilitating intercity economic links with the metropolis through transport, the cross-boundary innovation enclave allows the secondary cities to exploit the metropolitan agglomeration for technology and talent. The idea is that the attracted talent will serve manufacturing in secondary cities. As an interviewee from Quzhou, a manufacturing city in western Zhejiang province, argued: “Because it is more difficult for high-end talent to settle in Quzhou, it is taking such a space in Hangzhou to allow talent to stay there, but hopefully make contributions to Quzhou.” (Interview, 13 December 2021).

In summary, innovation enclaves have been developed in the hope of enhancing the role of secondary cities in innovation and entrepreneurship, by borrowing high-end intelligence and innovation resources from the more developed primary cities. The city-regional imaginary is achieved through the interaction between the two places, namely by luring the “phoenix” (more advanced manufacturing and technology as well as talent and entrepreneurs of the outer reaches of the metropolis) into the “nest” (of the secondary cities). This imaginary and associated strategizing highlights how spillovers do not occur spontaneously, but are orchestrated by proactive secondary cities. Indeed, some leading cities show hatred, or at least a conservative attitude, towards the behavior of secondary cities in their setting up enclaves in their jurisdictions, seeing it as *waqiangjiao* (the theft of technological and talent resources).

5. Conclusions

This study examines the modalities of city regionalism in China, through the lens of secondary cities. Most of the existing literature on city regionalism in China focuses on the role of nation-states or primary city governments and allied stakeholders, and there is a lack of research that unravels the agency of smaller regional and provincial – secondary cities. To fill this gap, this study draws attention to the city-regional imaginaries, orchestrated by secondary cities to take advantage of agglomeration and facilitate spillovers. By juxtaposing three types of city-regional imaginaries – cross-boundary new towns, cross-boundary industrial parks, and cross-boundary innovation enclaves – it articulates the agency and logic of secondary cities that underly metropolitan-centric and agglomeration-led city regionalism. As a result of these imaginaries, Chinese city regions are evolving into a megalopolitan or urban-regional corridor spaces that are not limited by administrative lines, but increasingly resemble the form extensively documented in the Western literature as a mosaic or archipelago of interconnected nodes competing and cooperating with each other (Scott, 2019).

Our case studies cover three different modes of city-regional imaginaries, which are mainly initiated by secondary cities. First, the cross-boundary new town imaginary shows a development path planned by secondary cities, to integrate functionally with the neighboring primary city. Jurisdictional boundaries are dissolved by way of transport connectivity (proximity) and service leveling (quality) which are common tactics used to attract businesses and populations. Second, the imagery of the cross-boundary industrial park is usually raised by secondary cities that are not adjacent to primary cities. Initially, the imaginary was designed to attract low-end industries from the metropolis to relocate to the secondary cities, thus freeing up space for the metropolis to upgrade.

Later, the imagination was modified through cooperation between secondary cities and successful development zones in the metropolis. Joint parks are expected to attract newly expanded production, or directly attract new enterprises by using the brand name and similar management and services of an original primary city zone. Third, the cross-boundary innovation enclave is commonly proposed out of frustration from the inability to attract and retain businesses and talent in secondary cities. Despite the lack of knowledge resources, these secondary cities seek to incubate start-ups and later facilitate mass production, by establishing innovation enclaves at the jurisdictional boundary with the primary city.

The case studies discussed in this paper reveal the concrete agencies and processes through which spillovers occur, from the perspective of secondary cities. Moving away from the assumption of common processes of agglomeration and spillover operating across city regions, this empirical research considers the diversity of profiles, trajectories, challenges, and opportunities that exist for a variety of cities that make up these regions. Moreover, the results implied that spillovers do not occur autonomously, as the agglomeration and trickle-down theories of city-regional development suggest (Rodríguez-Pose, 2008). This is particularly the case in the current dominant agglomeration approach to city regionalism, under the prevailing international competition to build global city-regions in particular. The selected cases based in China provide a rich and nuanced narrative of the different responses of smaller regional and provincial cities, and how these initiatives contribute to the formation of city, global city and megalopolitan regions. The case studies show that these secondary cities are not passive actors waiting for the trickle-down to occur. They offer an 'outside-in' theorization of city-regional development, rather than a simple conceptualization from the perspective of primary cities, where the city region is formed by waiting for the outflow of population and economic activity at a particular turning point. The evidence we have been able to present regarding the strategizing of Chinese city authorities with respect to cross-boundary developments, contrasts with the typically regressive dumping of locally unwanted land uses on neighboring authorities common in the west (Phelps, 2025). Instead, the Chinese evidence speaks directly to the urban and economic planning imagination apparent in patterns and processes of urbanization in urban peripheries (Phelps et al., 2025) and the agency of secondary city governments exert in the construction of an interplace economy within city systems and city regions in particular (Phelps, 2004, 2017, 2021a, 2021b).

This study identified several potential research agendas for future studies. First, this study demonstrates the importance of exploring the agency of secondary cities in city-regional formation. Attention to secondary cities is further justified by the growing call for a relational lens into city-regional studies (MacLeod & Jones, 2007). The relational perspective should not only refer to the global-local nexus, but also include equally complex dynamics within the city region. Second, and related, this study shows that the formation of city regions is not a spontaneous outcome of economic development. There is an absolute agency that distinguishes city-regional planning from planning city regions. Placing the study of planning city regions at the forefront of the research agenda focuses more on potential outcomes of city-regional development as processes of economization, which are shaped by the geopolitics of globalization and international relations, and also the geopolitical economy within the nation-state, where political dynamics between different levels of cities and other actors prevail (Jonas & Moiso, 2018). Thus, this analytical lens would help inform the society of small- and medium-sized cities, where the majority of the urban population lives, about the potential to be pathfinders, by pursuing innovative development pathways in areas such as well-being and catch-up development, and the potential to be political actors, backing relevant voices, lobbying power, and representation in contexts where some places matter more than others. Third, that being said, in this paper we have concentrated on elucidating the presence, logic and manifestation of secondary city strategizing but not on weighing its effects.

Throughout our discussion of the cross-boundary new towns, industry parks and innovation enclaves, the effects of intentional secondary city strategizing when set against the unintentional effects spillovers of primary cities remains unclear, and could usefully be investigated further. Fourth, comparative analysis could focus both on secondary city strategizing in the round and on strategizing with respect to jurisdictional boundaries in particular. Here Chinese conditions emerge not as incomparable to those in the west, but simply as those that extend the known spectrum of city system dynamics including specifically: modes of city, global city and megalopolitan region formation; the extent and depth of suburban and secondary city strategizing; and strategizing centred on jurisdictional boundaries.

CRediT authorship contribution statement

Yi Li: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nicholas A. Phelps:** Writing – review & editing, Conceptualization. **Guoliang Xu:** Validation.

Ethical statement

Given the fact that China has no equivalent system on ethics application, the collaborative research does not have signed document regarding human ethics approval. Nonetheless, we did state our purpose when we contacted the interviewees. Before we conducted our interviews, we asked the permit of our interviewees to record and also to cite some of their comments in our work in the future. We have obtained their consent to state their professional roles to make the quotation a bit robust. Our interviewees agreed with this as long as their name is anonymized.

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Declaration of competing interest

None.

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TBC

Data availability

No data was used for the research described in the article.

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